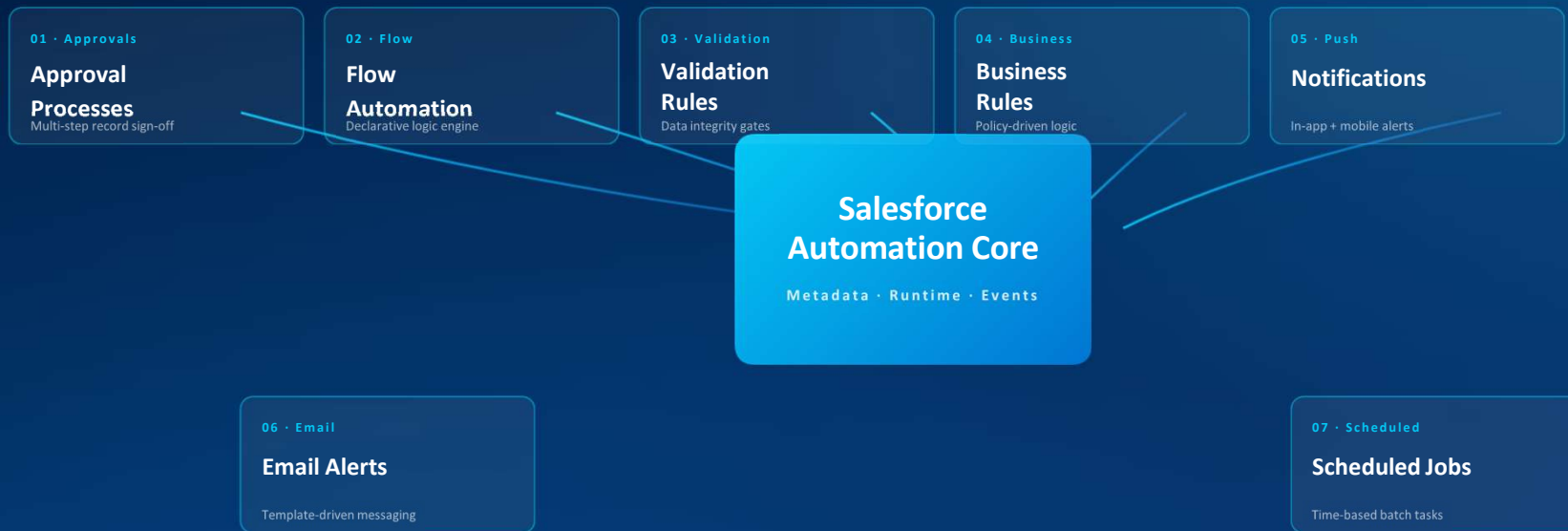


Automation Landscape

Seven automation building blocks orchestrated across the Salesforce platform



All automations execute on the Lightning Platform runtime, share governor limits, and are triggered by **DML events**, **time**, **user actions**, or **platform events**.

Approval Processes — Flow Chart

Multi-step, criteria-driven record approval with escalation and rejection paths



Key Configuration Components

Initial Submitters

Roles, groups, or users authorized to kick off approval

Approver Assignment

User field, hierarchy, queue, or related user

Field Updates & Actions

Auto-set Status, Approval Date, Approver Name

Recall & Reassign

Submitter or admin can withdraw / delegate

Flow Types & Runtime Architecture

Declarative automation engine — trigger source drives flow type

Flow Type Matrix

Screen Flow

User-facing wizards, guided data entry, embedded on Lightning pages

TRIGGER: User click / button / component

Record-Triggered Flow

Before/After Save on insert, update, delete — replaces Workflow & PB

TRIGGER: DML event on record

Scheduled Flow

Runs at fixed cadence over a batch of records

TRIGGER: Time schedule + entry query

Platform Event-Triggered

Reacts to published platform events for real-time integration

TRIGGER: PE message on event bus

Autolaunched Flow

Called from Apex, REST, process, or another flow

TRIGGER: Programmatic invocation

Flow Runtime · Block Diagram



Validation Rules & Business Rules

Where data integrity meets policy enforcement in the save path

Save Order — Where Rules Fire

- | | | |
|---|--|----------------|
| 1 | Load original record + submitted values | System |
| 2 | System validation (required, field type, layout) | System |
| 3 | Before-save flows (Record-Triggered) | Automation |
| 4 | Validation Rules run — halt save if any fails ⚠ | Data Integrity |
| 5 | Duplicate rules evaluated | System |
| 6 | Apex triggers + Business Rules (Bus. Logic) 🧠 | Policy Layer |
| 7 | Commit to database ✓ | Persist |

Validation Rules

Data Integrity

- Declarative formulas that must evaluate FALSE
- Block save + display custom error
- Field-level or page-level messages
- Fast, no code required
- Example: `Discount__c > 0.30 && !Approved__c`

Business Rules

Policy Logic

- Cross-object, multi-step decision logic
- Implemented via Flow, Apex, or Decision Table
- Can call external services, calculate values
- Governs process outcomes & routing
- Example: Route order to distributor based on postal code + SKU

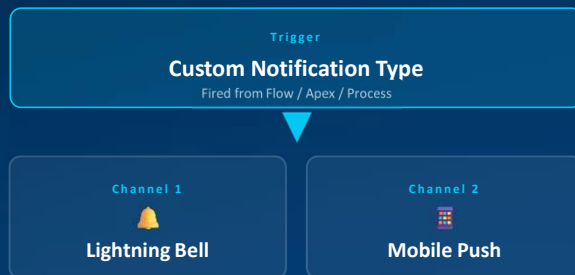
Notifications, Email Alerts & Scheduled Jobs

Three delivery mechanisms — event-driven, template-driven, time-driven

01 · Notifications

Push & In-App

Real-time bell + mobile push signals

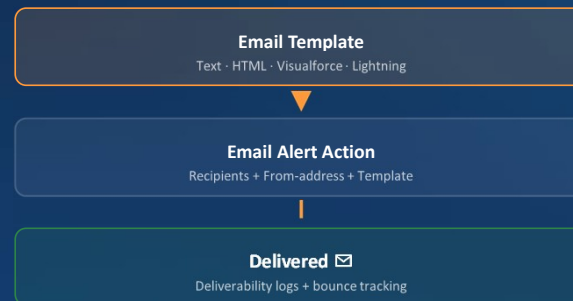


- Target: users, roles, or groups
- Payload: title, body, target record link
- Great for: approval requests, escalations, field visit reminders
- No inbox clutter — dismisses on read

02 · Email Alerts

Templated Email

Merge-field messages via Salesforce SMTP

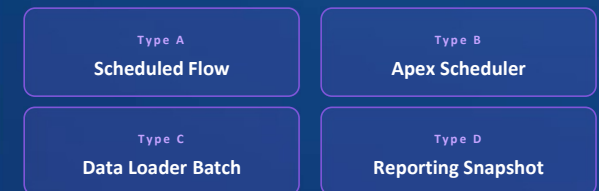


- Called by Flow, Approval, Process, Apex
- Governor limit: 5,000 mass emails / day / org
- Best for: order confirmations, quota alerts
- Use Marketing Cloud / Pardot for campaigns

03 · Scheduled Jobs

Time-Based Batch

Cron-style tasks for recurring operations



- Cron expression or hourly/daily/weekly UI
- Max 100 concurrent Apex scheduled jobs
- Monitor via Setup > Scheduled Jobs
- Common uses:
 - Nightly quota recalculation
 - Stale opportunity cleanup
 - Weekly distributor rollup
 - Territory realignment

Integration Landscape

Salesforce at the center of the enterprise data mesh — inbound & outbound wiring



Integration modes: Real-time (REST/SOAP/Streaming) · Near-real-time (Platform Events) · Batch (Bulk API/ETL)

Sales Cloud ↔ ERP — Wiring Diagram

Bi-directional data flows across the Order-to-Cash lifecycle



Common patterns: Outbound Message · Named Credentials + REST callout · Platform Events · Salesforce Connect (OData)